

Appendix: Active Inference, Quantum, and Type-Theoretic Notation

E. Active Inference and Cognitive Models

Term	Symbol	Definition	First Use
Active inference	—	Framework in which perception and action are unified as variational free energy minimization	§7
Active sampling	—	Agent's selection of actions to confirm or update case assignments via sensory evidence	§7
Belief	—	Bayesian	§7

F. Quantum and Topological Terms

Term	Symbol	Definition	First Use
Amplituhedron	—	Positive geometry encoding scattering amplitudes; connected to TQNN execution traces	§8
$\dagger$ -compact category	—	See $\dagger$ -compact closed category in §B	§8
CPTP map	—	Completely Positive Trace-	§8

G. Logical and Type-Theoretic Terms

Term	Symbol	Definition	First Use
β -reduction	—	Computational reduction step in λ -calculus; corresponds to cut elimination in proofs	§3
Church–Rosser property	—	Confluence of β -reduction: all reduction sequences converge to the same normal form	§3
Curry–	—	Correspondence	§3